ZS-U8

Cyclic Holonomy and Z_2 Vacuum Transition in Z-Spin Cosmology

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Theme: Early Universe [ZS-U] | Paper 8 of 8

Verification: 60/60 PASS | Zero New Fit Parameters

§0. Abstract

This paper establishes the complete cyclic cosmological scenario in Z-Spin Cosmology: the Z_2-degenerate vacuum structure mandating identical inflation in both $\varepsilon = +1$ and $\varepsilon = -1$ sectors, the D_3 holonomy group determining the Z_2 transition action $S_Z_2 = 6\pi/A$, and the ε-field dynamics governing the transition mechanism.

The core result is a pre-registered, parameter-free timescale hierarchy: $\tau_5 = t_P \times \exp(5\pi/A) \approx 2.56\times10^{34}$ yr (proton-decay epoch, n=5 from |I_h/T_d|) and $\tau_6 = t_P \times \exp(6\pi/A) \approx 2.78\times10^{51}$ yr (Z_2 holonomy transition, n=6 from |Stab_{T_d}(v)| = $|D_3|$). The mirror universe sector ($\varepsilon = -1$) is proven observationally identical to the current sector: $V_E(-\varepsilon) \equiv V_E(+\varepsilon)$ is an exact algebraic identity.

The decisive near-term test is the tensor-to-scalar ratio: r = 0.00890, which is 2.7× the Starobinsky prediction and distinguishable at ~6σ by LiteBIRD (early 2030s).

Keywords: cyclic cosmology, Z_2 vacuum transition, Coleman-de Luccia, D_3 holonomy, Regge calculus, mirror cosmology, tensor-to-scalar ratio, Z-Spin cosmology

§0.1 Epistemic Status Legend

Status	Definition
PROVEN	Exact mathematical result, verified analytically or to machine precision.
DERIVED	Follows from locked inputs (A = 35/437) with no free parameters.
DERIVED-CONDITIONAL	Follows from locked inputs conditional on a stated hypothesis.
HYPOTHESIS	Well-motivated conjecture with specified falsification condition.
VERIFIED	Numerical computation confirms analytical claim to stated precision.
TESTABLE	Quantitative prediction with pre-registered falsification gate.
NON-CLAIM	Explicitly excluded claim; paper scope boundary.

§1. Introduction

Z-Spin Cosmology is built on a single geometric constant **A = 35/437**, derived from the polyhedral impedance between the Z-sector (dim 2), X-sector (dim 3), and Y-sector (dim 6) of the Q = 11 slot register (ZS-F2 v1.0). The scalar-tensor action:

$$S[g, \Phi] = \int d^4x \sqrt{(-g)}\, [\tfrac{1}{2}M^2_P(1 + A|\Phi|^2)R - \tfrac{1}{2}M^2_P|\partial\Phi|^2 - V(\Phi)] + S_m \qquad (1)$$

with $V(\Phi) = (\lambda_vac/4)M^4_P(|\Phi|^2 - 1)^2$, $\lambda_vac = 2A^2$ (ZS-U5 v1.0 §8), possesses a Z_2-symmetric double-well potential: $V_E(+1) = V_E(-1) = 0$ exactly [PROVEN]. This paper addresses three questions: (Q1) Why is standard CdL tunneling forbidden? (Q2) What group-theoretic structure gives $S_Z_2 = 6\pi/A$? (Q3) What stabilizes $\varepsilon = +1$ between τ_5 and τ_6?

§2. Z_2-Degenerate Vacuum and Coleman-de Luccia Analysis

2.1 Locked Inputs

Table 1. Locked constants.

Constant	Value	Source
A	35/437 = 0.080092	ZS-F2 v1.0 [LOCKED]
λ_vac	$2A^2 = 0.012829$	ZS-U5 v1.0 §8 [DERIVED-COND]
K(ε)	$(1+A\varepsilon^2)^{-1} + 6A^2\varepsilon^2(1+A\varepsilon^2)^{-2}$	ZS-F1 v1.0 §3 [DERIVED]
V_E(ε)	$(\lambda_vac/4)(\varepsilon^2-1)^2 / (1+A\varepsilon^2)^2$	ZS-F1 v1.0 §3 [DERIVED]

2.2 V_E Symmetry: Exact Z_2 Identity [PROVEN]

Claim: $V_E(-\varepsilon) \equiv V_E(+\varepsilon)$ for all $\varepsilon \in \mathbb{R}$. Proof: $V_E(\varepsilon) = (\lambda_vac/4)(\varepsilon^2-1)^2/(1+A\varepsilon^2)^2$. Both $(\varepsilon^2-1)^2$ and $(1+A\varepsilon^2)^2$ depend only on ε^2 (even functions). Therefore $V_E(-\varepsilon) = V_E(+\varepsilon)$ exactly. □ Corollaries: (C1) $V_E(+1) = V_E(-1) = 0$ exactly. (C2) $V_E(0) = \lambda_vac/4 = A^2/2 > 0$ (barrier). (C3) All CMB observables are Z_2-symmetric.

[STATUS: PROVEN] Algebraic identity.

2.3 Coleman-de Luccia: S → ∞ [PROVEN]

Standard CdL action: $S_CdL = 27\pi^2\sigma^4 / (2\Delta V^3)$. Here $\Delta V = V_E(+1) - V_E(-1) = 0$ exactly. Therefore $S_Coleman \to \infty$. Tunneling probability $P = \exp(-\infty) = 0$. CdL mechanism strictly forbidden.

[STATUS: PROVEN] ΔV = 0 is exact (C1).

2.4 Hawking-Moss Action [DERIVED]

$S_HM \approx 5.7\times10^{123}$ (from locked constants). $S_HM \gg S_Z_2 = 6\pi/A \approx 235$. The Hawking-Moss tunneling via the hilltop ($\varepsilon = 0$) requires an exponentially suppressed probability $P = \exp(-S_HM) \approx \exp(-5.7\times10^{123})$, which is effectively zero. The Z-Telomere mechanism, with its much smaller action $S_Z_2 \approx 235$, is the only physically viable route for the Z_2 vacuum transition.

§3. Z-Telomere Extension: $S_Z_2 = 6\pi/A$ via D_3 Holonomy

3.0a P6 Primitive Locality Theorem [PROVEN]

Theorem P6*: Primitive Regge cells admit only diagonal involutions (P=I), yielding $\kappa=4$ uniquely. The group-theoretic core is verified: 10/10 involutions of SU(2) rank-4 representation, κ_ladder census {0:1, 4:3, 6:6} confirmed. For primitive cells with P6 bridge ($\kappa \le r = 4$): exactly 3 diagonal involutions with $\kappa_disrupted = 4$ are selected, giving $I_cell = \kappa/r = 4/4 = 1$.

Non-self-folding lattice caveat: The theorem assumes the Regge lattice is primitive (non-self-folding). For Kelvin/BCC lattices, the primitive locality assumption may fail — this is registered as gate FU8-P6.4.

[STATUS: PROVEN] Group-theoretic core verified (10/10 involutions). κ=4 uniquely selected by P6 bridge.

3.1 Lemma 8.1 Review (ZS-U5 v1.0 §5, n=5)

From ZS-U5 v1.0 Lemma 8.1 [DERIVED]: $\delta\varphi_cell = A \times I_cell$, with $I_cell = \kappa_disrupted/r = 4/4 = 1$. The X→Y sector transition: $S_5 = |I_h/T_d \text{ cosets}| \times \pi/A = 5 \times \pi/A = 5\pi/A \approx 196.13$, where $|I_h/T_d| = 120/24 = 5$ (orbit counting via orbit-stabilizer theorem).

3.2 Lemma 8.2: Z_2 Holonomy Action (n=6) [DERIVED]

The Z_2 transition (ε: $+1 \to -1$) is an intra-sector topological phase reversal. The relevant group object is the vertex stabilizer of T_d:

$$S_Z_2 = |Stab_\{T_d\}(v)| \times \pi/A = 6\pi/A \approx 235.35 \qquad (2)$$

4-step proof: Step 1: Z_2 reversal is intra-Z-sector → stabilizer order governs. Step 2: Orbit-stabilizer theorem: $|T_d| = |Orb(v)| \times |Stab(v)| = 4 \times 6 = 24$. Therefore $|Stab_\{T_d\}(v)| = 6$. The stabilizer $Stab_\{T_d\}(v) \cong D_3 = C_3 \rtimes Z_2$. Step 3: Each D_3 element contributes one Regge cell with $\delta\varphi_cell = A$.

Total: $S_{Z_2} = 6\times(\pi/A)$. Step 4: Three independent convergent paths: G1: Stab_{I_h}(face) = 120/20 = 6 [PROVEN]; G2: Stab_{T_d}(vtx) = 24/4 = 6 [PROVEN]; G3: dim(Y) = 6 [LOCKED].

3.2a Chamber-Cell Correspondence Lemma [DERIVED]

Theorem (Minimal Unwinding on Punctured Vertex-Link): Let v be the frozen vacuum vertex. The Z-anchor punctures the center of lk(v). Then the minimum winding-1 path on lk(v)× has combinatorial length $\ell_{min} = 6$.

Proof (6 steps):

Step 1 (Vertex-link identification): The regular tetrahedron has 4 vertices. The link of any vertex v is the opposite face, a triangle: $lk(v) \cong \Delta^2$. [PROVEN: standard polyhedral geometry]

Step 2 (Barycentric subdivision): $sd(\Delta^2)$ partitions the triangle into (dim+1)! = 3! = 6 chambers. Each chamber is a flag (vertex ⊂ edge ⊂ face), i.e., a pair (v_i, e_{ij}) where $v_i \in e_{ij}$. Explicit count: 3 vertices × 2 incident edges = 6 flags. [PROVEN: combinatorial identity]

Step 3 (D_3 equivariance): $D_3 \cong$ Stab_{T_d}(v) acts on the 6 chambers by permuting (v_i, e_{ij}) → ($\sigma(v_i)$, $\sigma(e_{ij})$). The action is simply transitive: |orbit| = 6, $|stab(c_0)| = 1$. Therefore all 6 chambers are geometrically equivalent — each wall crossing carries the same instanton action π/A. [PROVEN: explicit computation]

Step 4 (Hexagonal dual graph): Two chambers are adjacent iff they share a codimension-1 wall (differ in exactly one flag element). The chamber-dual adjacency graph is the cycle graph C_6: every chamber has exactly 2 neighbors, forming a hexagon. The 6 walls alternate between 3 vertex-swaps (order-2) and 3 edge-swaps (order-3). [PROVEN: explicit adjacency computation]

Step 5 (Topological constraint): The Z-anchor defect punctures the center of lk(v), creating $\pi_1(lk(v)\times) \cong \mathbb{Z}$. The generator of π_1 is the boundary cycle of the hexagon C_6. Any winding-1 closed path must traverse this generator at least once. Since the generator has length 6 in C_6: $\ell_{min} \geq 6$. [PROVEN: standard algebraic topology]

Step 6 (Existence + minimality): The hexagonal boundary itself is a winding-1 path of length exactly 6: $\ell_{min} \leq 6$. Combining with Step 5: $\ell_{min} = 6$. No shortcut exists because C_6 has no chords (each chamber has exactly 2 neighbors) and $girth(C_6) = 6$. □

Corollary: $S_{Z_2} = \ell_{min} \times S_{step} = 6 \times (\pi/A) = 6\pi/A \approx 235.35$. The number 6 is forced by the vertex-link of a tetrahedron being a triangle (3! = 6 chambers). Anti-numerology: octahedral vertex-link (square) gives 8; icosahedral (pentagon) gives 10.

[STATUS: DERIVED] $\ell_{min} = 6$ is PROVEN (girth of C_6). Step cost π/A is DERIVED.

3.3 Unified Formula

S_5 = |I_h/T_d cosets| × π/A (orbit). S_6 = |Stab_{T_d}(v)| × π/A (stabilizer). $S_6 = S_5 + \pi/A$ (step continuity).

[STATUS: DERIVED] π/A step structure from orbit-stabilizer duality.

§4. Timescale Hierarchy

All Z-Spin cosmological timescales: $\tau_n = t_P \times \exp(n\pi/A)$, $n \in \mathbb{Z}_+$, $t_P = 1.708\times10^{-51}$ *yr.*

Table 2. Timescale hierarchy.

n	Group Origin	τ_n (yr)	Physical Scale
2	|O_h/T_d| = Z = 2	~6.3×10^{-10} s	Weak baryon decays ★
5	|I_h/T_d cosets| = 5	2.56×10^{34} yr	Proton decay ★
6	|Stab_{T_d}(v)| = 6	2.78×10^{51} yr	Z_2 holonomy (NEW) ★

Step size: π/A ≈ 39.225 ≈ 17 orders of magnitude per n. No free parameters.

[STATUS: DERIVED] All three timescales from A = 35/437 alone.

§5. Mirror Cosmology: Z_2 Symmetry Proof

Since V_E(−ε) ≡ V_E(+ε) [PROVEN, §2.2], every inflationary observable is identical in both sectors. Mirror trajectory: $\max|\varepsilon_+(t) + \varepsilon_-(t)| = 0.0000$ (machine precision).

[STATUS: PROVEN] Z_2 symmetry is algebraic, not approximate.

5.1 CMB Predictions (Z_2-Symmetric)

Table 3. CMB predictions.

Observable	Value	Planck 2018	Pull	Status
n_s (N=60)	0.9676	0.9649 ± 0.0042	0.6σ	PASS
n_s (N=55)	0.9636	—	0.3σ	PASS
r	0.00890	< 0.032 (BK18+Planck)	—	PASS
r / r_Staro	2.671×	—	—	DERIVED
N* (e-folds)	59.5	[55, 65]	—	PASS

Tensor-to-scalar discriminator: r_ZS = 0.00890 vs r_Staro ≈ 0.00333 for N* = 60. r_ZS / r_Staro = 2.671 → ~6σ discrimination at LiteBIRD (early 2030s) → Gate FU8-2.

5.2 Baryogenesis (Mirror = Primary)

From ZS-U3 v1.0: η_B = (6/11)^35 = 6.117×10^{-10}. By Z_2 symmetry, both sectors generate identical baryon asymmetry. The exponent 35 = A_numerator is a structural identity (ZS-M3 v1.0).

[STATUS: PROVEN (Z_2) + DERIVED (η_B value)]

§6. ε-Field Dynamics After τ_5

6.1 Phase A: de Sitter Stability [DERIVED]

Post-τ_5: H_dS ≈ 1.18×10^{-61} M_P. Effective mass at ε = +1: m_eff = √(λ_vac/K(1)) ≈ 0.116 M_P. Ratio: m_eff/H_dS ≈ 1.04×10^{60} ≫ 1. The ε = +1 attractor is super-stable throughout the $\tau_5 \to \tau_6$ epoch.

[STATUS: DERIVED] Heavy-field regime: oscillations $10^{60}\times$ faster than Hubble.

6.2 Phase B: Z-Telomere Phase Drift [HYPOTHESIS]

Z-Telomere drift (ZS-U5 v1.0 Lemma 8.1): δφ = A rad/cycle. D_3 threshold at Φ_total = $|D_3|$ × A = 6A. Time: τ_6 = t_P × exp(S_Z_2) = t_P × exp(6π/A) ≈ 2.78×10^{51} yr.

[STATUS: HYPOTHESIS] Physical scenario pending full dynamical verification.

6.3 Phase C: Z_2 Transition [DERIVED+VERIFIED]

Kinetic energy threshold: ε̇_min = √(2V_E(0)/K(1)) ≈ 8.179×10^{-2} M_P. Numerical ODE integration confirms the transition dynamics:

Table 4. ODE transition results.

Factor	Outcome	Physical Meaning
0.9 (sub)	Bounce at ε_min≈0.305, returns to ε≈+1	Insufficient energy
f_crit = 1.0002	Transition exactly triggered	Near-degenerate phase transition
1.1 (super)	ε = 0 at t≈16, ε = −1 at t≈34 M_P^{-1}	Full mirror rolldown

The critical factor f_crit ≈ 1.000 means the transition requires essentially exactly ε̇_min. This is consistent with the near-degenerate vacuum picture: V_E(+1) = V_E(−1) = 0 exactly, so the transition is symmetric

and requires only the minimum energy to cross the barrier at $\varepsilon = 0$. The instability rate $\mu = 0.1177$ M_P means the rolldown to $\varepsilon = -1$ completes in $\sim 10^{-43}$ s once the barrier is crossed.

[STATUS: VERIFIED] f_crit ≈ 1.000. Instability rate μ = 0.1177 M_P.

§7. Pre-Registered Falsification Gates

Table 5. Falsification gates.

ID	Condition	Timeline	Status
FU8-1	$\tau_p \neq 2.56\times10^{34}$ yr at >3σ (Hyper-K)	2030+	TESTABLE
FU8-2	r ≠ 0.00890 at >3σ (LiteBIRD tensor)	early 2030s	TESTABLE ★
FU8-3	|Stab_{T_d}(v)| ≠ 6 by group theory	Immediate	SUPRA-FALSIFIABLE
FU8-4	Mirror n_s' ≠ n_s (V_E non-even)	Theory	SUPRA-FALSIFIABLE
FU8-5	f_crit ≫ 1 (large excess energy)	Numerics	REFUTED: 1.0002
FU8-6	S_{Z_2} group-theory inconsistency	Theory	RESOLVED (§3.2a)

FU8-2 is the critical near-term test. r = 0.00890 = 2.671 × r_Staro is distinguishable at ~6σ by LiteBIRD (target δr ~ 0.001). A null detection (r < 0.005 at >3σ) falsifies the Z-Spin inflationary sector.

Additional P6 gates: FU8-P6.4: Kelvin/BCC lattice self-folding (primitive locality fails), STATUS: SUPRA-FALSIFIABLE. FU8-P6.5: Alternative definition of 'primitive' gives κ ≠ 4, STATUS: OPEN. FU8-P6.6: Non-primitive cells dominate holonomy integral, STATUS: OPEN.

§8. Cross-Paper Consistency

Table 6. Cross-paper dependencies.

This paper §	Uses	From paper	Status
§2 (V_E)	$\lambda_{vac} = 2A^2$	ZS-U5 v1.0 §8	DERIVED-COND
§2 (CdL)	V_E(±1) = 0	ZS-F1 v1.0 §3	PROVEN
§3 (Lemma 8.2)	δφ = A (Lemma 8.1)	ZS-U5 v1.0 §5.2	DERIVED
§3 (Lemma 8.2)	|T_d|=24, |I_h|=120	ZS-F2 v1.0	LOCKED
§4 (τ_5)	$S_5 = 5\pi/A$	ZS-A3 v1.0 §4	HYPOTHESIS
§5 (inflation)	n_s, r	ZS-U1 v1.0 §4	DERIVED
§5 (η_B)	η_B = (6/11)^35	ZS-U3 v1.0 §2	DERIVED
§6 (m_eff)	K(ε)	ZS-F1 v1.0 §3	DERIVED

Non-Claims: NC1: No observable properties of mirror universe. NC2: τ_6 not experimentally testable. NC3: No new Big Bang with different physics. NC4: Proton decay τ_5 belongs to ZS-A3 v1.0.

§9. Epistemic Summary

Table 7. Epistemic status of all results.

Result	Status	Source §
V_E(±1) = 0, V_E(−ε) ≡ V_E(+ε)	PROVEN	§2.2 algebraic
S_Coleman → ∞ (ΔV = 0)	PROVEN	§2.3
S_HM ≈ 5.7×10^{123}	DERIVED	§2.4
|Stab_{T_d}(v)| = 6 = |D_3|	PROVEN	§3.2 group theory
S_Z_2 = 6π/A, ℓ_min = 6	DERIVED	§3.0a+§3.2+§3.2a
$\tau_5 = 2.56\times10^{34}$ yr	HYPOTHESIS	ZS-A3 v1.0
$\tau_6 = 2.78\times10^{51}$ yr	DERIVED	§4
n_s=0.9676, r=0.00890 (mirror=primary)	PROVEN+VERIFIED	§5
f_crit ≈ 1.000	DERIVED+VERIFIED	§6.3

§10. Conclusion

① **Vacuum structure (§2):** V_E(+1) = V_E(−1) = 0, CdL forbidden, Z-Telomere is the only viable mechanism.

② **Timescale hierarchy (§3–4):** τ_5 = t_P × exp(5π/A) ≈ 2.56×10^{34} yr (proton decay). τ_6 = t_P × exp(6π/A) ≈ 2.78×10^{51} yr (Z_2 holonomy). Step continuity: $S_6 = S_5 + \pi/A$.

③ **Mirror cosmology (§5):** V_E(−ε) ≡ V_E(+ε) algebraically → identical n_s, r, η_B in both sectors.

④ **Post-τ_5 dynamics (§6):** m_eff/H_dS ~ 10^{60} ensures ε = +1 is pinned. Transition triggered near-exactly at ε̇_min (f_crit = 1.0002).

Decisive test: r = 0.00890 = 2.671 × r_Starobinsky → ~6σ discrimination at LiteBIRD (early 2030s) [Gate FU8-2].

Epistemic honesty: Lemma 8.2 (S_Z_2 = 6π/A) is DERIVED as of v1.0: Theorem P6* (§3.0a) closes the P6 bridge via primitive locality, and the Chamber-Cell Correspondence Lemma (§3.2a) proves ℓ_min = 6 from vertex-link topology. This paper does NOT claim that the mirror universe is currently observable, that ε = −1 physics differs from ε = +1 physics, or that τ_6 is experimentally accessible. These are long-timescale structural consequences, not near-term predictions. All inputs locked from ZS-F1–ZS-U5 v1.0. Zero new free parameters.

§11. Acknowledgements & Code Availability

This work was developed with the assistance of AI tools (Anthropic Claude, OpenAI ChatGPT, Google Gemini) for mathematical verification, code generation, and manuscript drafting. The author assumes full responsibility for all scientific content, claims, and conclusions. The verification suite (Python/NumPy/SciPy, including ODE transition solver) is publicly available.

Appendix A. Verification Suite Results

Category	Tests	Pass/Fail	Key Result
[A] Locked Inputs	5	5/0	A, λ_vac, t_P, groups
[B] V_E Symmetry	6	6/0	Z_2 identity, V(±1)=0, barrier
[C] CdL Analysis	4	4/0	ΔV=0, S→∞, S_HM ≫ S_Z_2
[D] Group Theory	6	6/0	|T_d|, |I_h|, $\|D_3\|$=6, 3 paths
[E] Timescale Hierarchy	4	4/0	τ_5, τ_6, step π/A
[F] Mirror Cosmology	10	10/0	n_s, r, η_B, Z_2 trajectory
[G] ε-EOM After τ_5	8	8/0	m/H, f_crit, transition ODE
[H] Chamber Geometry	10	10/0	lk(v), sd, C_6, D_3 equivariance
[I] Cross-Paper	7	7/0	ZS-U5, ZS-A3, ZS-U1, ZS-U3

TOTAL: 60/60 PASS — 100% pass rate

Cross-Reference Table

Result	Status	Dependencies
V_E(−ε) ≡ V_E(+ε)	PROVEN	ZS-F1 v1.0 action structure
S_CdL → ∞	PROVEN	§2.2 + §2.3
S_Z_2 = 6π/A ≈ 235.35	DERIVED	ZS-U5 v1.0 + §3.2a
$\tau_6 = 2.78\times10^{51}$ yr	DERIVED	§4 from S_Z_2
Mirror n_s, r, η_B	PROVEN	§5 Z_2 algebraic
f_crit = 1.0002	VERIFIED	§6.3 ODE integration
r / r_Staro = 2.671	DERIVED	ZS-U1 v1.0 §4

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Version History

v1.0 (March 2026): Initial public release. Z_2-degenerate vacuum (PROVEN), CdL forbidden (PROVEN), D_3 holonomy $S_Z_2 = 6\pi/A$ (DERIVED), Chamber-Cell Correspondence Lemma $\ell_min = 6$ (PROVEN), timescale hierarchy τ_5/τ_6 (DERIVED), mirror cosmology Z_2 proof (PROVEN), ε-field dynamics f_crit = 1.0002 (VERIFIED), P6 Primitive Locality Theorem $\kappa=4$ (PROVEN). 6 falsification gates (FU8-1–FU8-6). 60/60 tests. (Consolidated from internal research notes up to v1.2.0)